

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD) "Coding Requirements" section:

FR-DLT-001: Load Customer and Order Data into Delta Tables

Title: Load Customer and Order Data into Delta Tables

Description: The system must read source CSV data from a specified volume and load it into Delta tables for customer and order data.

Preconditions:

Source CSV files are available at the specified locations:

`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`

- Delta tables for customer and order data do not exist or are empty

Main Flow / Functional Steps:

1. Read customer CSV data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` into a Delta table named "customer".
2. Read order CSV data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata` into a Delta table named "order".
3. Add a new column "TotalAmount" to the "order" Delta table, calculated as the product of "PricePerUnit" and "Qty" columns.

FR-DLT-002: Clean and Process Customer and Order Data

Title: Clean and Process Customer and Order Data

Description: The system must remove null and duplicate records from the customer and order Delta tables.

Preconditions:

- Customer and order Delta tables exist and contain data

Main Flow / Functional Steps:

1. Remove rows with null values from the "customer" and "order" Delta tables.
2. Remove duplicate rows from the "customer" and "order" Delta tables.

FR-DLT-003: Create and Load Order Summary Table

Title: Create and Load Order Summary Table

Description: The system must create an "ordersummary" table if it does not exist and load joined customer and order data into it using SCD type 2.

Preconditions:

- Customer and order Delta tables exist and contain clean data
- "ordersummary" table does not exist or is empty

Main Flow / Functional Steps:

1. Create the "ordersummary" table in the specified catalog and schema if it does not exist.
2. Join the "customer" and "order" Delta tables on the "CustId" column.
3. Load the joined data into the "ordersummary" table using SCD type 2.

FR-DLT-004: Update Order Summary Table on Customer Change

Title: Update Order Summary Table on Customer Change

Description: The system must update the "ordersummary" table when there is a change in the customer table.

Preconditions:

- "ordersummary" table exists and contains data
- Customer table has been updated

Main Flow / Functional Steps:

1. Identify changes in the "customer" table.
2. Update the "ordersummary" table by making old records inactive and new records active.
3. Update the "StartDate" and "EndDate" columns accordingly to maintain history.

FR-DLT-005: Create and Load Customer Aggregate Spend Table

Title: Create and Load Customer Aggregate Spend Table

Description: The system must create a "customeraggregatespend" table if it does not exist and load aggregated data from the "ordersummary" table.

Preconditions:

- "ordersummary" table exists and contains data
- "customeraggregatespend" table does not exist or is empty

Main Flow / Functional Steps:

1. Create the "customeraggregatespend" table in the specified catalog and schema if it does not exist.
2. Aggregate the "TotalAmount" column from the "ordersummary" table by "Name" and "Date" columns.
3. Load the aggregated data into the "customeraggregatespend" table.

FR-DLT-006: Implement Delta Live Tables**Title: Implement Delta Live Tables**

Description: The system must implement the above steps using Delta Live Tables.

Preconditions:

- All previous steps have been completed successfully

Main Flow / Functional Steps:

1. Implement the data pipelines using Delta Live Tables to perform the above steps.